

7 Describing the trend

Iteration	e_1	e_2	m_1	m_2	b_1	b_2
0 (Start)	-10	-85	-1	2	1	1
1	-9.49	-78.23	-0.9814	1.9657	1.0001	1.00085
2	-9.05	-72.33	-0.8693	1.9363	1.0002	1.0016
3	-8.66	-67.12	-0.8128 -0.761	1.9109	1.0003	1.0024
4 (Final)	-0.761	-1.889	-0.761	1.889	1.0004	1.0031

Computing the cost function $J = \frac{1}{2} (e_1^2 + e_2^2)$ at the start and end confirms this trend numerically:

$$J_0 = \frac{1}{2} (-10^2 + -85^2) = \frac{1}{2} (100 + 7225)$$

$$J_0 = 3662.5$$

$$J_3 = \frac{1}{2} (-8.66^2 + -67.12^2) = \frac{1}{2} (75 + 4505.1)$$

$$J_3 = 2290.05$$

Conclusion: Parameters are moving to reduce error, all parameters values are moving in directions that reduce prediction error.

Evidence:- errors e_1 and e_2 are decreasing in magnitude

- The cost J is also decreasing confirming that the gradient descent algorithm is moving toward lower error
- Parameters are adjusting systematically

Therefore: This confirms that the gradient descent algorithm is working correctly and the use of a small learning rate was ~~to~~ because of the large magnitudes.